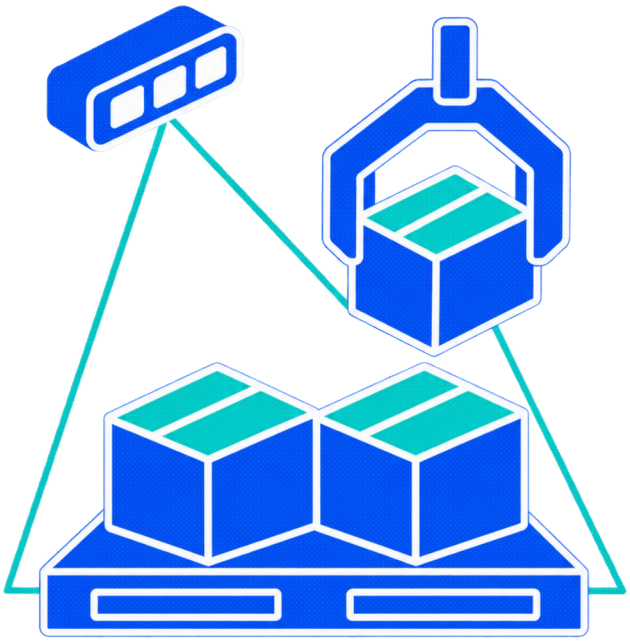
MRDVS® 迈尔微视



AW3 PalletEye

Depalletizing Solution White Paper

RGB and depth fusion for bag and box pick poses

Zhejiang MRDVS Technology Co., Ltd.

V1.0 · 2026-11-11

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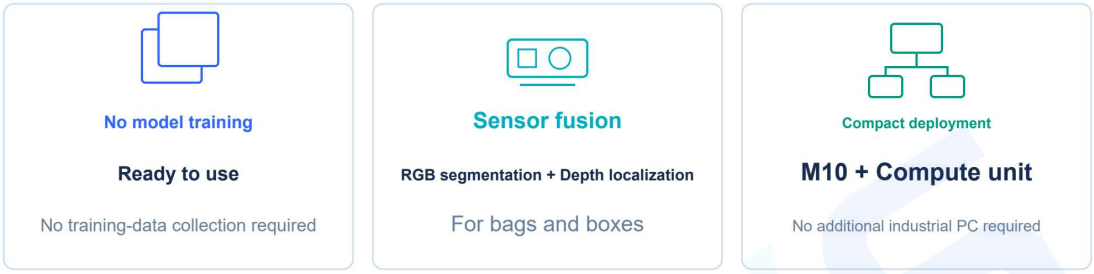
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1 Value and Applications

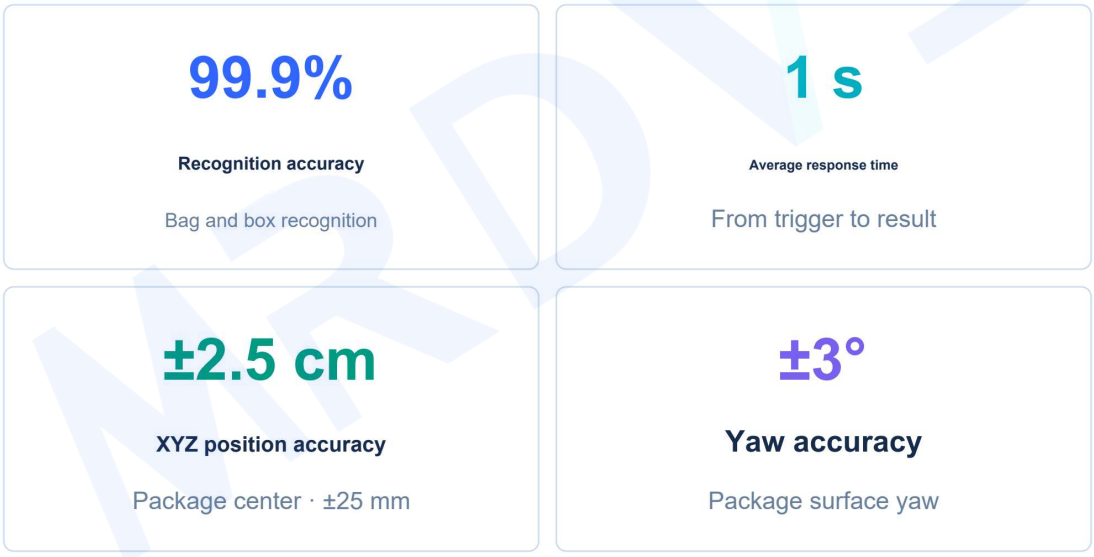
PalletEye supports item-by-item picking in production and warehousing. It detects regularly stacked bags and boxes and provides the robot with package positions, surface orientations and pick order.

Applications include raw-material feeding, production-line loading and outbound handling. The M10 camera and compute unit form the vision system; the robot and end effector are selected for the process.

1.1 Key Advantages



1.2 Performance at a Glance



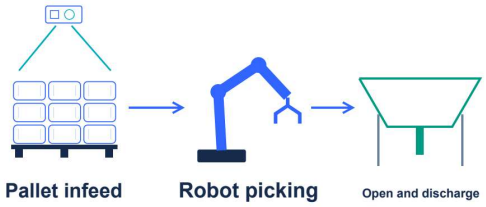
Average response time is 1 s from trigger to result. Robot picking and transfer time are additional.

1.3 Applications

From incoming pallets to downstream processes, PalletEye supplies the visual localization required for item-by-item robotic picking.

Raw-Material Feeding

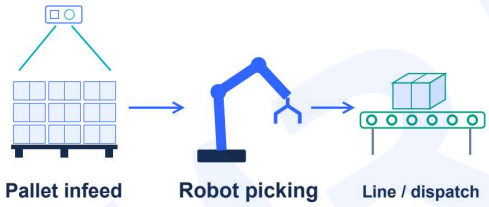
Pick bags from a pallet and deliver them to an opening or discharge station for downstream processing.





Production and Warehouse Supply

Pick individual packages from a full pallet onto a conveyor for production-line loading or outbound truck loading.

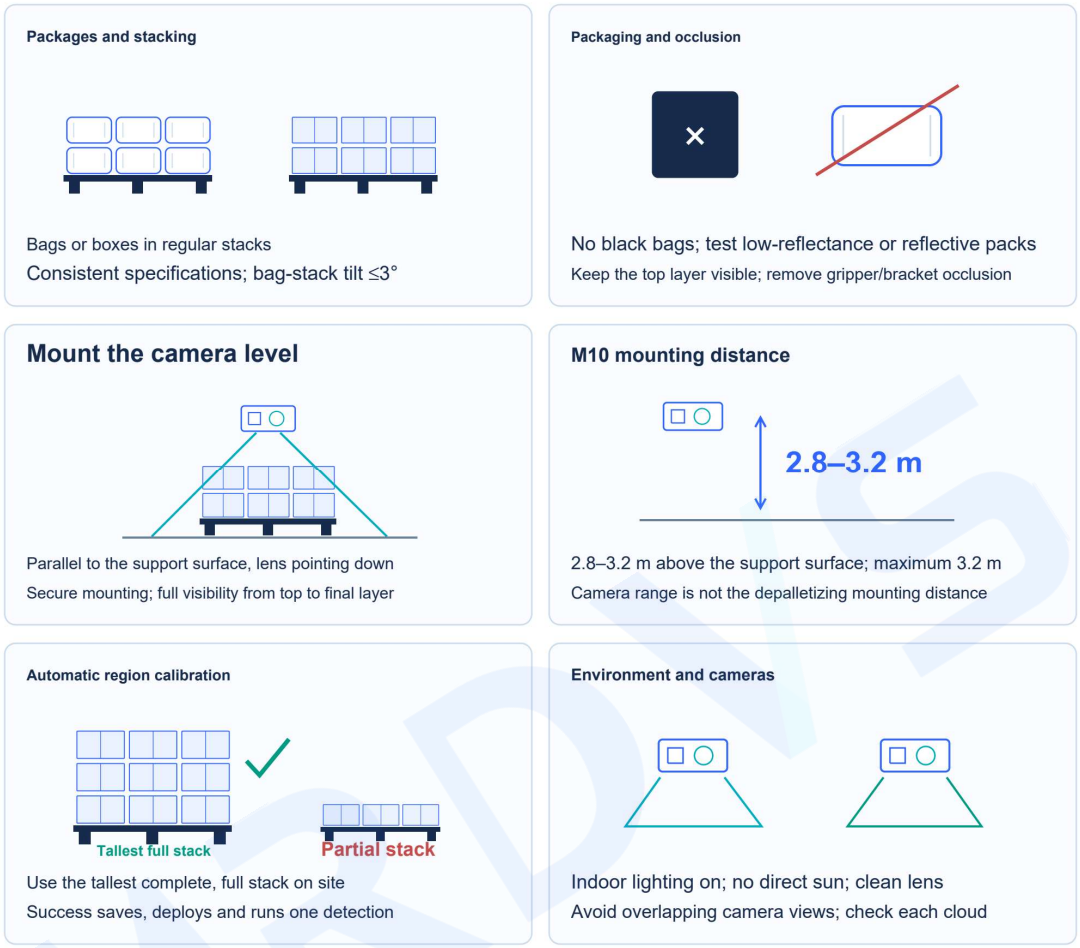




Left: workflow illustration. Right: rendered application concept. Equipment and line layout depend on the project.

1.4 Packages and Operating Conditions

Supports bags and boxes in regular patterns, including five-bag interlocking, ring, six-bag and eight-bag arrangements. Package specifications must be consistent within each stack.

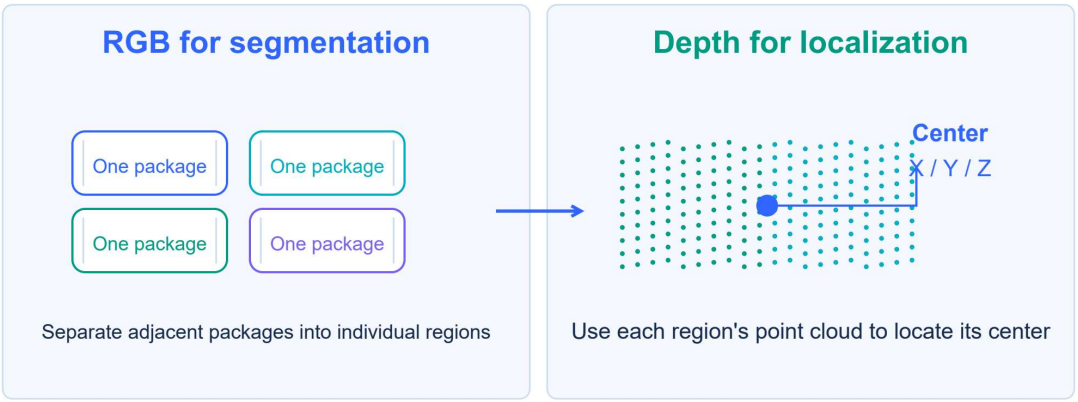


> Automatic region calibration requires the tallest complete, full stack used on site. Partial or shorter stacks can produce inaccurate bounds or fail because the region is too small.

Evaluate severely deformed, damaged, deeply dented or open packaging with samples first. Use actual packages to verify recognition, camera coverage and picking performance.

2 Recognition and Localization

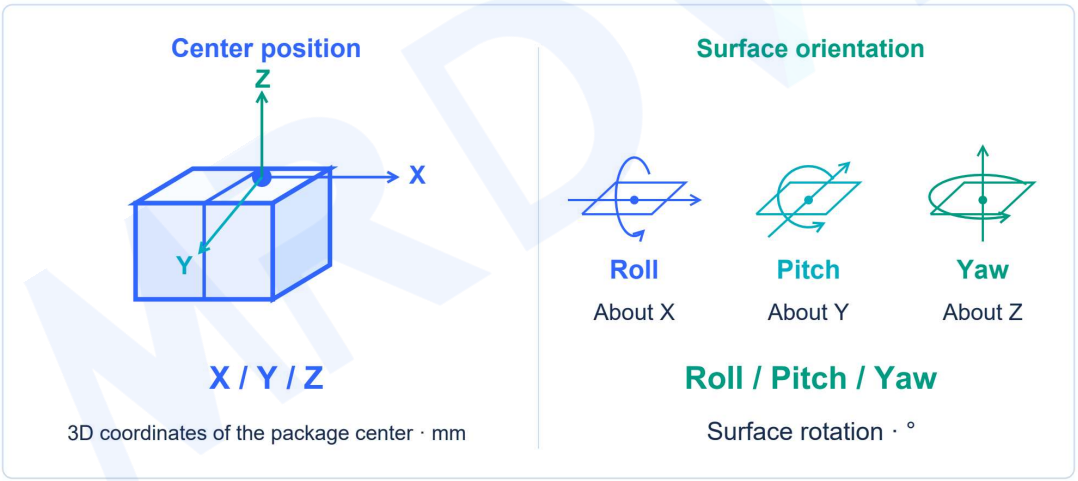
2.1 Segmentation and Center Localization



Segment packages → Match point clouds → Locate centers

RGB images separate individual packages; depth clouds locate their centers. Within the work area, size and layer checks select top-layer targets, and in-layer rules determine pick order.

2.2 Center Position and Surface Orientation



X, Y and Z locate the package center in mm. Roll, Pitch and Yaw describe surface rotation about the X, Y and Z axes respectively, in degrees.

3 Supporting Hardware

M10 captures RGB and depth data. The compute unit runs the depalletizing algorithm on Ubuntu Linux and serves as the AW3 center.



M10 RGBD camera

RGB and depth capture



Compute unit (center)

Ubuntu Linux · Algorithm processing

M10 camera and compute unit (product images, not to scale)

M10 RGBD Camera Specifications

Parameter	Specification
Model	M10
Sensor type	dToF RGB-D
Laser wavelength	940 nm
Output formats	Depth / IR / RGB images / 3D point cloud
ToF resolution and frame rate	480 × 320 @ 5 fps
ToF field of view (H × V)	90° × 60° (±3°)
RGB resolution and frame rate	1920 × 1080 @ 5 fps
RGB field of view (H × V)	90° × 60° (±3°)
Distance range	0.5–5 m (10% reflectance)
Ranging accuracy	±30 mm
Average power consumption	≤4 W
Dimensions / weight	80 × 37 × 25 mm / approx. 190 g
Power supply	12–27 V
Communication interface	Ethernet
Ingress protection / eye safety	IP54 / CLASS I
Operating temperature	−20°C to 60°C
Storage temperature	−40°C to 85°C
Time synchronization	PTP
Software support	C/C++/ROS SDK
Supported operating systems	Windows 7/8/10/11, Linux, Arm Linux/ROS
Ambient-light immunity	100 kLux

Camera distance range and ranging accuracy are hardware specifications; they do not define depalletizing mounting height or actual picking accuracy.

4 Deployment and Configuration

4.1 Two Operating Methods



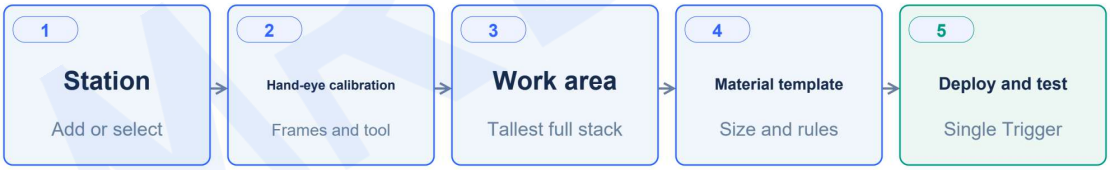
Method 1: Connect a monitor to the compute unit via HDMI and operate its installed software directly.



Method 2: Install AW3 on a Windows industrial PC or laptop and connect to the compute unit over the network.

The algorithm runs on the compute unit in both cases. Connect the camera, compute unit and optional PC to the same LAN through a switch.

4.2 On-Site Setup and Validation

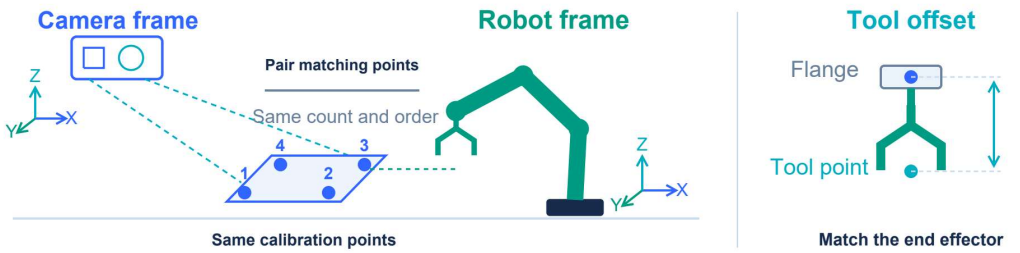


For each station, complete hand-eye calibration, work area setup and material template configuration, then test with Single Trigger. Use the tallest complete, full stack for automatic region calibration; success automatically saves, deploys and runs one detection.

> Before first use, confirm authorization and deploy the Soft Pack Depalletizing camera parameters in AW3. For connections, authorization and software operation, see the AW3 User Manual.

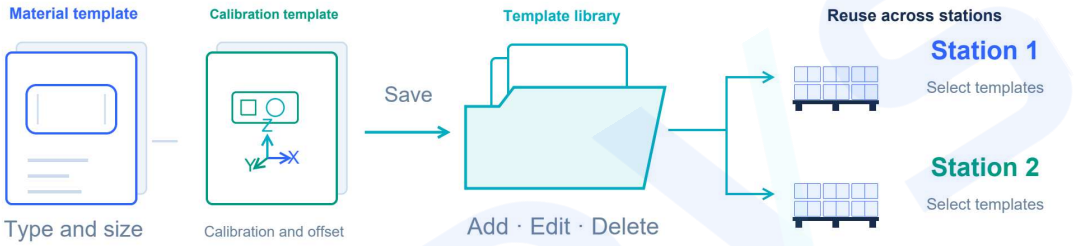
4.3 Configuration and Management Features

4.3.1 Hand-Eye Calibration and Tool Offsets



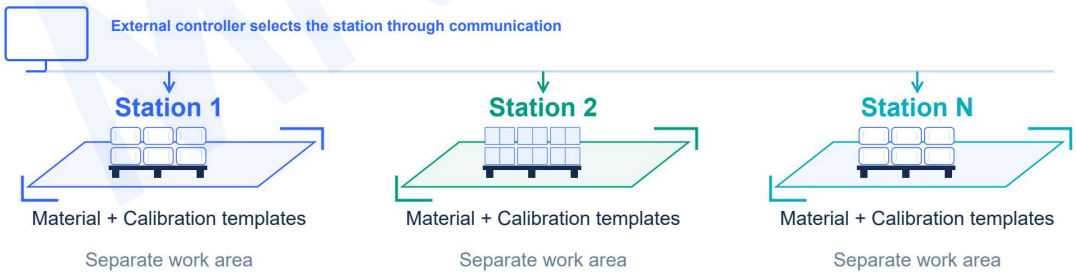
Hand-eye calibration relates the camera and robot coordinate frames; tool offsets adapt the output to the end effector. Obtain camera points by marker detection or manual entry, then pair each with the corresponding robot point.

4.3.2 Reusable Material and Calibration Templates



Material templates store type, dimensions and recognition rules. Calibration templates store calibration parameters and tool offsets. Both can be added, edited, deleted and selected by multiple stations.

4.3.3 Multiple Stations and Remote Switching



Create a station for each working position, with its own material template, calibration template and work area. An external system selects the station through communication to switch configurations.

Models and Function Parameters are shared within a camera group. Material templates, calibration templates and work areas are selected or configured per station.

5 Project Evaluation and Integration

Provide package, site and control-system information before evaluation to establish suitability and hardware requirements.

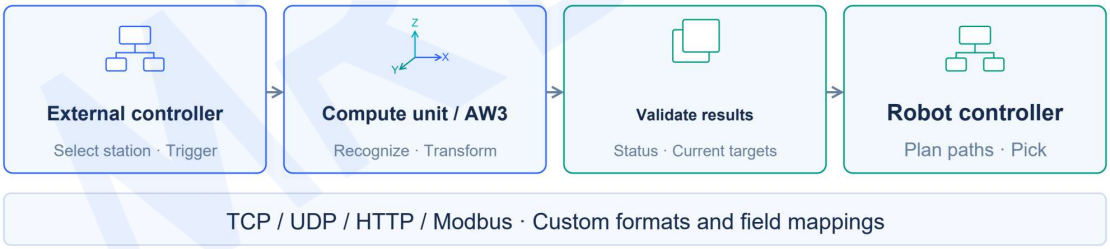
5.1 Information for Evaluation

Category	Recommended information
Packages and stacking	Packaging type, size range, stacking pattern, maximum stack height, and photos or samples.
Site and equipment	Station layout, mounting space, lighting, robot model and end-effector type.
Process and interfaces	Feeding or loading process, target machine cycle time, protocol and required result fields.

5.2 Sample Testing and System Integration



Test actual packages at maximum full-stack height, intermediate layers and the final layer. Verify recognition, localization, picking and overall cycle time, then complete communication integration.



Supports TCP, UDP, HTTP and Modbus with customizable formats and field mappings. The vision system supplies recognition and localization results; the robot controller plans paths and executes picks.

For operating procedures and result fields, see the Depalletizing Algorithm User Manual.

6 Selection FAQs

Question	Answer
Do we need to collect samples and train a model?	Common bags and boxes do not require collecting training data first. On-site template setup, calibration, region configuration and testing with actual packages are still required.
How are bags and boxes configured?	Create separate material templates with appropriate dimensions and recognition rules. Keep package specifications consistent within each stack.
Is a separate industrial PC required?	No. Connect a monitor to the compute unit via HDMI for local operation, or install AW3 on a Windows PC and connect remotely.
How many cameras can one compute unit support?	Up to two cameras in parallel. Select the number and placement to cover the stations.
Can the system serve multiple stations?	Yes. Each station has its own material template, calibration template and work area. Switch configurations through communication.
How does it connect to an existing robot system?	Use TCP, UDP, HTTP or Modbus. Configure six-axis poses and message fields to match the receiving controller.
Does the 1 s response time include robot handling?	No. It is the average time from trigger to vision result. The machine cycle also includes picking, transfer and process waiting time.
How can we confirm suitability for our site?	Provide the information in Chapter 5 and validate actual samples with the planned mounting layout and end effector.

Revision History and Technical Support

Version	Date	Changes
V1.0	2026-11-11	Product capabilities, applications, recognition principles, hardware deployment and validation requirements.

Technical Support

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Knowledge Community: https://docs.mrdvs.com/en

Visit MRDVS on GitHub